Exp 9a – Solutions and Dilutions

INTRODUCTION

In this experiment, you will make two solutions. One solution will be made by dissolving a solid solute (salt, NaCl). The other solution will be made by diluting a small amount of the first solution with water to make a new solution with new concentration. You will then perform various calculations to express the concentration of salt in both solutions in different units.

BACKGROUND

Bodies of water around the world can be classified as fresh water, brackish water, saline water or briny water depending on the amount of salt dissolved in them. Saline solutions (made with table salt, NaCl) have many uses in medicine including nasal washes, eye drops, and IV therapy. Brines are solutions that vary from somewhat concentrated to saturated. Because many bacteria do not grow well in high concentrations of sodium chloride, brines are occasionally used as food preservative (pickling).

In this experiment, you will prepare two sodium chloride (NaCl) solutions of varying concentrations. You will first mix sodium chloride and water to prepare a more concentrated "stock" solution and then you will make one "working" solution by diluting the stock solution.

There are various units of concentration that can be used to describe the amount of salt in each solution:

$$Molarity\ (M) = \frac{mol\ solute}{L\ solution} \tag{1}$$

$$Mass\ Percent\ \left(\frac{m}{m}\%\right) = \frac{mass\ solute}{mass\ solution} \times 100 \tag{2}$$

$$Molality\ (m) = \frac{mol\ solute}{kg\ solvent} \tag{3}$$

$$Mass\ Volume\ Percent\ \left(\frac{m}{v}\%\right) = \frac{mass\ solute\ in\ g}{mL\ solution} \times 100 \tag{4}$$

Additionally, the density of each solution can be calculated by:

$$Density\ Solution = \frac{mass\ solution}{volume\ solution} \tag{5}$$

SAFETY EQUIPMENT REQUIRED FOR LAB

- Goggles
- Lab coat

SUPPLY LIST

- Sodium chloride (solid)
- 10-mL volumetric pipet
- 50-mL volumetric flask
- 25-mL volumetric flask
- Pipet bulb
- Permanent marker
- Wash bottle
- One small stopper
- One large stopper

CLASSROOM SUPPLY AND REACTANT LIST

- Sodium chloride (solid)
- Food coloring

PROCEDURE

Make Solution "A"

1. ☐ Use a permanent marker (Sharpie®) to label a 50-mL volumetric flask "A."
2. ☐ Record the precise mass of the 50-mL volumetric flask with stopper (empty) on a balance.
3. ☐ Use a piece of weigh paper to obtain between 4.5-6.0 g of NaCl.
4. ☐ Pour the NaCl into the 50-mL volumetric flask. It is okay if some spills on the countertop. Spills can be cleaned up by sweeping the salt into the trash can.
5. ☐ Record the precise mass of the 50-mL flask (with stopper) now that it contains salt.
6. ☐ Use a water bottle (filled with DI water) to fill the flask approximately halfway with water. Add 1 drop of food coloring. The food coloring is to make your solutions visually different.
7. ☐ With the stopper on, swirl the flask until the NaCl is mostly dissolved then fill to the line with DI-Water and mix by inverting at least 10 times.
8. ☐ If needed, carefully add more water so that the bottom of the meniscus is at the line.
9. ☐ Record the precise mass of the 50-mL flask (with stopper) now that it contains "Solution A".

Make Solution "B"

1. ☐ Use a Sharpie® to label a 25-mL volumetric flask "B."
2. ☐ Record the precise mass of the 25-mL volumetric flask with stopper (empty) on a balance.
3. ☐ Use a 10-mL volumetric pipet to transfer 10.00 mL of "Solution A" to the flask for "Solution B".
4. ☐ Use a water bottle (filled with DI water) to fill the flask to the line with water.
5. ☐ With the stopper on, swirl the flask and mix by inverting at least 10 times.
6. ☐ Record the precise mass of the 25-mL flask (with stopper) now that it contains "Solution B".

CLEAN-UP AND WASTE DISPOSAL

1. ☐ After checking your data with your instructor, all solutions can be poured down the drain.
2. ☐ Flush volumetric pipets thoroughly with DI water from wash bottle.
3. ☐ Wash volumetric flasks with soapy water. Rinse with DI water.
4. ☐ Rinse the inside of the volumetric flasks and pipets with a small amount of acetone, to help them dry faster.
5. ☐ Remove any permanent marker using acetone.
6. ☐ There may be drying racks available for the volumetric flasks. If drying racks are available, place the volumetric flasks upside down in the rack. If no rack is available, place the equipment back in the original location.

CALCULATIONS

- Refer to the definitions of concentration units found in the 'Background' section of the experiment.
- It is recommended that you complete the calculations in the order presented in the data report, even though this is not the same order found in the Results table.
    - Rounded values should not be used in calculations. Write down the calculator output, and store values in your calculator so that you have the full number.
    - Round values for significant figures when reporting your answers in the Results table.
- Volumetric flasks and pipets are very precise. When considering significant figures in calculations, use the precise volumes below:

	Precise Volume
50-mL Volumetric Flask	50.00 mL
25-mL Volumetric Flask	25.00 mL
10-mL Volumetric Pipet	10.00 mL

DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

* Show your work and keep track of significant figures. **DATE _____**

PARTNER _____

DATA

	Solution A
Mass of 50-mL volumetric flask + stopper (empty)	
Mass of 50-mL volumetric flask + stopper + NaCl	
Mass of 50-mL volumetric flask + stopper + "Solution A"	

	Solution B
Mass of 25-mL volumetric flask + stopper (empty)	
Mass of 25-mL volumetric flask + stopper + "Solution B"	
Volume of "Solution A" used to make "Solution B"	

CALCULATIONS

Show your mathematical work, complete with units.

1. Mass of solute in Solution A:

2. Mass of solution for Solution A:

3. Mass of solvent for Solution A:

4. Moles of solute in Solution A:

5. Molarity (M) of Solution A:

6. Mass % (m/m%) of Solution A:

7. Molality (*m*) of Solution A:

8. Mass-Volume % of Solution A:

9. Density of Solution A:

10. Moles of solute in 10.00 mL of Solution A ("Moles of Solute" for Solution B):

11. Mass of solute used to make Solution B:

12. Mass of solution for Solution B:

13. Mass of solvent for Solution B:

14. Molarity (M) of Solution B:

15. Mass % (m/m%) of Solution B:

16. Molality (*m*) of Solution B:

17. Mass-Volume % of Solution B:

18. Density of Solution B:

RESULTS

	Solution A	Solution B
Mass Solute		
Mass Solvent		
Mass Solution		
Moles of Solute		
Molarity (M)		
Mass Percent (m/m %)		
Mass-Volume Percent (m/v %)		
Molality (*m*)		
Density of Solution		